MANAGEMENT INFORMATION SYSTEM

UNIT – III

Strategic and Project Planning for MIS

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Unit at a glance

1. General Business Planning **2.** Appropriate MIS Response **3.** MIS Planning – General **4.** MIS Planning – Details **5.** Conceptual Design – Definition of the Problem **6.** System Objective & System Constraints **7.** Analysis of Information Sources **8.** Alternative System Design & Selection of Optimal System **9.** Conceptual System Design Document

1 General Business Planning

Planning means deciding in advance what to do, how to do it, when to do it and who will do it. **Business planning** sets the organisation's direction. Because MIS exists to serve the business, the MIS plan must always grow out of the business plan.

1.1 Steps of Business Planning

1. **Set the mission and objectives** – why the organisation exists and what it wants to achieve.
2. **Analyse the environment** – strengths, weaknesses, opportunities and threats (SWOT).
3. **Formulate strategies** – the broad ways of reaching the objectives.
4. **Make policies and action plans** – detailed programmes, budgets and schedules.
5. **Implement and control** – carry out the plans and compare results with targets.

1.2 Levels of Planning

Level	Done by	Time span	Focus
Strategic planning	Top management	Long term (3–5+ years)	Mission, growth, new markets, overall direction
Tactical planning	Middle management	Medium term (1 year)	Allocating resources and implementing strategy by department
Operational planning	Lower management	Short term (days / months)	Day-to-day schedules and routine tasks

2 Appropriate MIS Response

An **appropriate MIS response** means that the MIS plan answers the needs of the business plan – it provides the right information, systems and resources to help the organisation reach its goals, no more and no less.

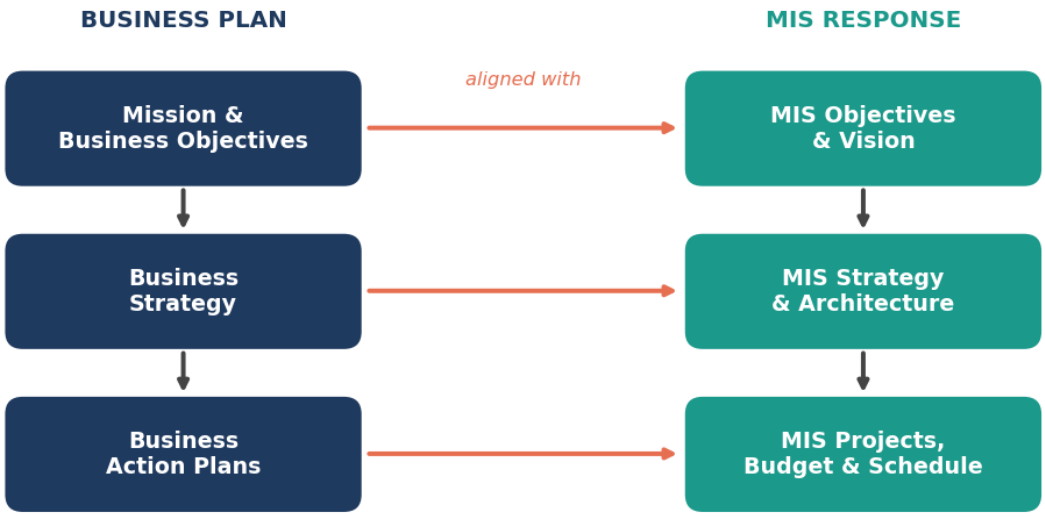


Fig 1: MIS plan aligned with the business plan

- MIS objectives must **support the business objectives**, not run separately from them.
- The **MIS strategy** decides which systems, technology and data are needed to support the business strategy.
- **MIS projects** are prioritised by their value to the business and by cost and risk.
- The MIS plan is **reviewed whenever the business plan changes**.
- Benefits: avoids wasted spending, ensures top-management support and gives a competitive advantage.

3 MIS Planning – General

General MIS planning gives the **overall picture** of how an MIS is planned. The aim is to study what the organisation needs and then prepare, resource and run a suitable MIS.

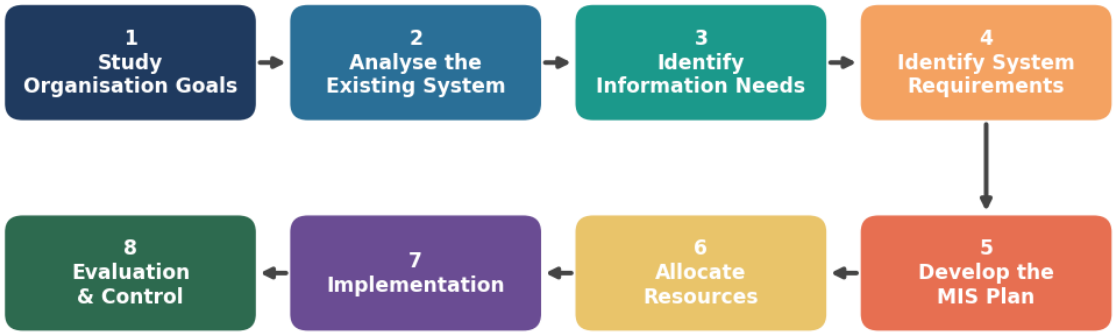


Fig 2: General MIS planning process

Step	Description
1. Study organisation goals	Understand the mission, objectives and plans of the business
2. Analyse the existing system	Find strengths, weaknesses and gaps in the current information system
3. Identify information needs	Find out what information each manager and department requires

Step	Description
4. Identify system requirements	Decide what the new system must do – hardware, software, data, users
5. Develop the MIS plan	Prepare a written plan with objectives, design, schedule and cost
6. Allocate resources	Assign money, people, equipment and time to the plan
7. Implementation	Install, test and start using the system; train the users
8. Evaluation & control	Check results against the plan and make corrections

4 MIS Planning – Details

Detailed MIS planning breaks the general steps into **more specific planning activities**, adding separate planning for resources, implementation and security.

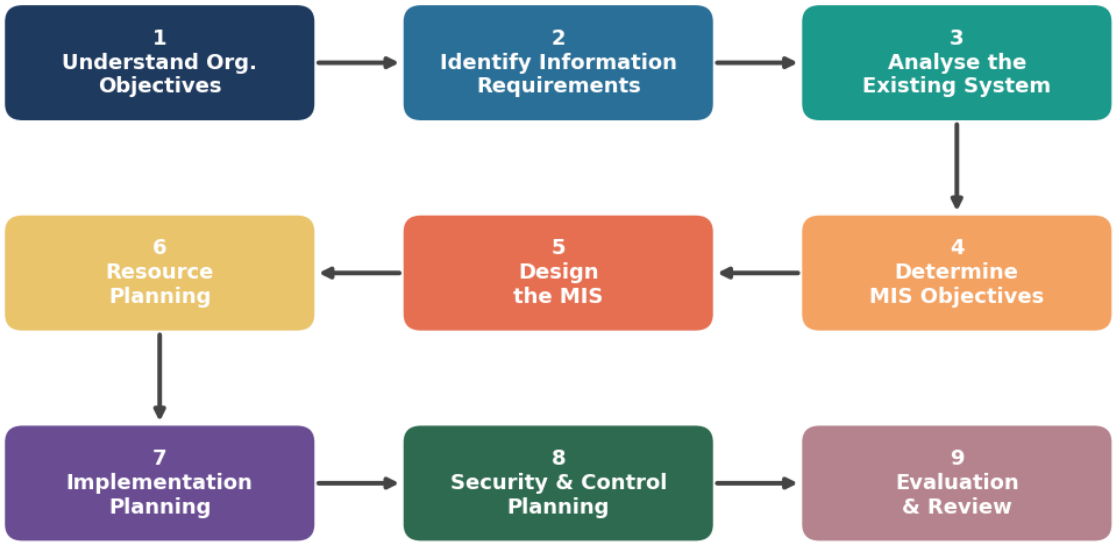


Fig 3: Detailed MIS planning process

Step	What it involves
1. Understanding organisational objectives	Study business goals, strategy and the structure of the organisation
2. Identify information requirements	List the information needed at strategic, tactical and operational levels
3. Analysis of the existing system	Review current systems, data flow, problems and their cost
4. Determining MIS objectives	Set clear, measurable goals for what the MIS must achieve
5. Designing the MIS	Prepare the overall design – inputs, processing, outputs, database
6. Resource planning	Plan hardware, software, staff, budget and time
7. Implementation planning	Plan installation, testing, training, conversion and schedule
8. Security & control planning	Plan access rights, backups, audit and protection against misuse
9. Evaluation & review	Plan how performance will be measured and the MIS improved

General vs Detailed MIS planning

General = 8 broad steps (study → plan → resource → implement → control).

Detailed = 9 specific steps; adds *determining MIS objectives*, *designing the MIS*, and separate *resource*, *implementation* and *security & control* planning.

5 Conceptual Design

Conceptual design is the first stage of designing an MIS. It gives a **broad, high-level picture** of the new system – what it must do and in what general way – without going into technical details. These details come later, in detailed design.

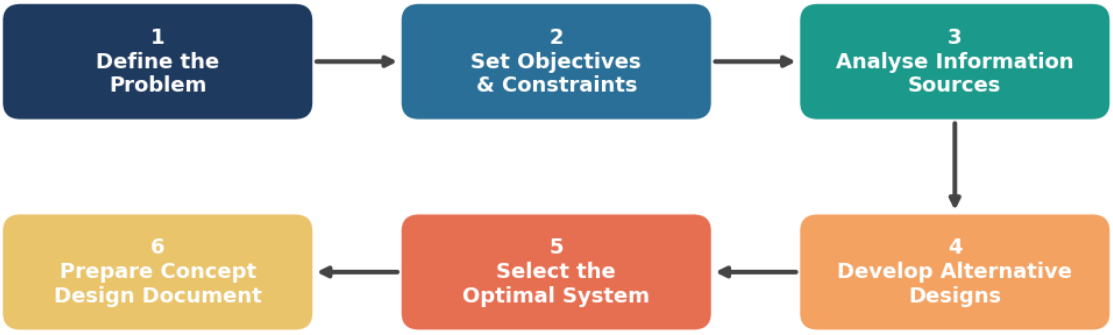


Fig 4: Steps of conceptual design

5.1 Definition of the Problem

The problem must be **clearly defined** before a solution is designed. A badly defined problem leads to a wrong or useless system.

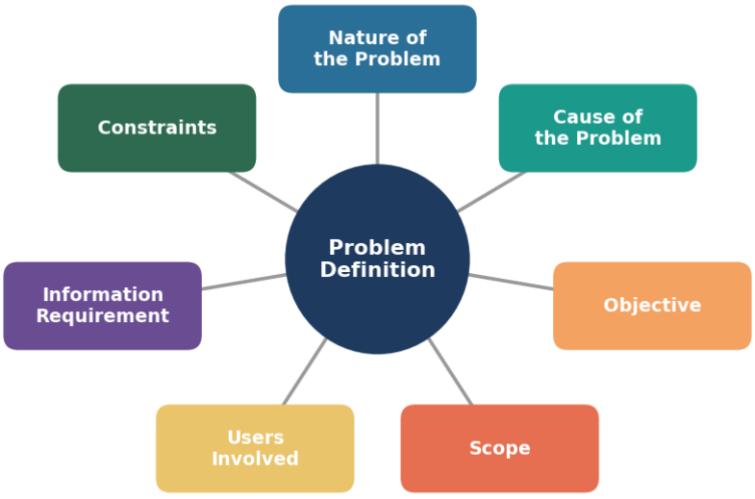


Fig 5: Elements of problem definition

Element	Question it answers
Nature of the problem	What exactly is wrong or needed?

Element	Question it answers
Cause of the problem	Why does the problem exist?
Objective	What should the new system achieve?
Scope	Which areas, departments and functions are included – and which are not?
Users involved	Who will use or be affected by the system?
Information requirement	What information must the system provide?
Constraints	What limits the solution – cost, time, technology etc.?

5.2 System Objective & System Constraints

The **system objective** states what the system is meant to achieve. A good objective is **clear, measurable, achievable and linked to the business goals** – for example, "reduce billing time from 2 days to 2 hours".

System constraints are the limits within which the system has to be designed. The common constraints are:

Constraint	Meaning	Example
Cost	Limited budget	Cannot spend more than the approved amount
Time	Fixed deadline	System must be ready before the new financial year
Organisational	Structure, policy, culture	Departments resist sharing data
Technical	Limits of available technology	Old hardware cannot run new software
Human resource	Skill and staff shortage	Few trained operators available
Legal	Laws and regulations	Data-protection and tax rules must be followed
Security	Need to protect data	Sensitive data needs restricted access

5.3 Analysis of Information Sources

The designer must find **where the required information will come from**. Sources of information are of two types:

Internal sources	External sources
Come from inside the organisation	Come from outside the organisation
Accounting and sales records, inventory data, personnel records, production reports, previous MIS reports	Customers, suppliers, competitors, government publications, market surveys, trade journals, internet and databases
Easy to obtain, cheaper and more reliable	May be costly and less certain, but vital for strategy

- Check each source for **availability, accuracy, cost and frequency** of the information.
- Decide whether data will be **collected, purchased or generated** by the system.

5.4 Alternative System Design

Usually more than one design can meet the objective. The designer prepares **alternative designs** and compares them. Each alternative is described by:

Feature	Details to decide
Hardware & software requirements	Which computers, servers, networks, operating system and packages

Feature	Details to decide
System architecture	Centralised, distributed or cloud-based structure
Processing methods	Batch, online (real-time) or a mix of both
Database design	What data is stored, how it is organised and shared
Input & output methods	How data enters the system and how reports are produced
Cost	Development, operation and maintenance cost
Time required	Time to develop and implement
Flexibility	Ease of future changes and growth
Security	Protection of data and system

5.5 Selection of the Optimal System

The alternatives are compared on fixed **selection criteria**, and the **best overall system** (the optimal system) is chosen.

Criterion	What is checked
Cost	Is it within the budget?
Benefits	What savings and gains will it give?
Technical feasibility	Can it be built with available technology?
Operational feasibility	Will users accept it and can the organisation run it?
Time	Can it be ready in time?
Reliability	How dependable is it?
Flexibility	Can it adapt to future needs?
Security	Is data adequately protected?

Illustrative example – weighted scoring (scores out of 10; made-up numbers to show the method):

Criterion	Weight %	Design A	Design B	Design C
Cost	20	8	6	5
Benefits	20	6	9	8
Technical feasibility	10	9	7	6
Operational feasibility	10	8	8	6
Time	10	9	6	5
Reliability	10	6	9	8
Flexibility	10	5	8	9
Security	10	6	8	9
Weighted total	100	7.1	7.6	6.9

✓ **Result of the example**

Design B has the highest weighted total, so it would be selected as the optimal system.

5.6 Conceptual System Design Document

The conceptual design is written down in a **Conceptual System Design Document**. It is shown to management for approval and becomes the base for detailed design.

No.	Content	Description
1	Problem definition	Statement of the problem and its causes
2	System objective	What the system must achieve
3	System scope	Boundaries – what is and is not covered
4	System constraints	Cost, time, technical and other limits
5	Information requirements	Information needed by each user
6	Analysis of information sources	Internal and external sources identified
7	Alternative system designs	The designs considered
8	Selection of optimal system	The chosen design and reasons for choosing it
9	Major inputs & outputs	Main data entering and reports leaving the system
10	Implementation considerations	Plan, resources, training and risks for implementation

★ Quick Revision & Exam Questions

Key points in one glance

- **Business plan → MIS plan:** MIS objectives, strategy and projects must match the business plan.
- **General MIS planning (8 steps):** goals → existing system → information needs → system requirements → MIS plan → resources → implementation → evaluation & control.
- **Detailed MIS planning (9 steps)** adds MIS objectives, design, and separate resource, implementation and security planning.
- **Conceptual design:** problem → objectives & constraints → information sources → alternative designs → optimal system → design document.
- **Constraints:** Cost, Time, Organisational, Technical, Human resource, Legal, Security.
- **Selection criteria:** Cost, Benefits, Technical & Operational feasibility, Time, Reliability, Flexibility, Security.

Likely questions

1. Explain general business planning and the appropriate MIS response.
2. Describe the steps of general MIS planning.
3. Explain detailed MIS planning with a diagram.
4. What is conceptual design? Explain the steps involved.
5. How is the problem defined in conceptual design?
6. Explain system objectives and the common system constraints.
7. Discuss internal and external sources of information.
8. Explain alternative system design and the criteria for selecting the optimal system.
9. What is a conceptual system design document? List its contents.